Southern New Hampshire University

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this system is to help **DriverPass** provide better driver training solutions.
- The system will allow students to:

  - Take online practice exams
  - Schedule driving lessons
  - Access training materials
  - The system will also support internal staff (IT, secretary, management) in managing operations efficiently.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass identified a gap in the market: more than 65% of students fail their driving test because they only studied previous tests without getting proper training.
- The system needs to provide online practice tests, online class content, and the ability to book on-the-road driving sessions to address this gap.
- DriverPass offers three training packages that include different combinations of driving hours, in-person lessons, and online class access.
- The system must be cloud-based and accessible from any computer or mobile device. DriverPass does not want to handle backup or security infrastructure.
- The system should also integrate with the DMV to receive updates on rules, policies, and sample test questions, so the content stays current.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Provide students with access to online practice exams and training content
- Allow users to schedule, modify, and cancel driving lessons
- Enable staff to manage users, appointments, and reports
- Maintain secure and accurate data tracking
- Ensure DMV updates are integrated into the system
- Deliver a cloud-based, accessible system

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- System must be web-based, cloud hosting and accessible from desktop and mobile devices.
- Pages should load fast
- Real time updates when online
- Support multiple users when concurrent without any issues.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- Run on web-browser (chrome, edge, Firefox) and mobile devices.
- Cloud infrastructure
- Database
- The owner must have the ability to download reports

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system must distinguish between different user roles (owner, IT manager, secretary, driver, customer) and provide the necessary access for each.
- Track user activity
- Ensure accurate scheduling

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The IT officer must be able to add, remove, and modify user accounts without changing the systems code.
- The owner should be able to enable or disable training packages without needing a developer.
- The system must adapt to DMV updates and policy changes.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users must log in with: Username and password.
- The system must use secure, encrypted connections (HTTPS) to protect data exchange between client and the servers. Specially with credit card information to avoid leaks.
- If a user forgets their password, the system should allow for an automatic password reset process.
- If a user has multiple failed attempts to log in, the account should be locked to prevent brute force attacks.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow users to create and manage accounts
- The system shall allow customers to schedule driving lessons
- The system shall allow customers to cancel or modify reservations
- The system shall assign drivers and vehicles to appointments
- The system shall track lesson duration and scheduling (2-hour sessions)
- The system shall allow users to select training packages
- The system shall allow admins to disable packages
- The system shall provide online practice exams
- The system shall track test progress (status, score, time)
- The system shall store driver notes and lesson details
- The system shall generate activity reports
- The system shall allow password reset functionality
- The system shall integrate DMV updates
- The system shall process customer payments

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface must be web-based and accessible from desktop and mobile browsers.
- Customers need pages for: account registration, login, profile management, package selection, reservation scheduling, online class and practice test access, and a contact page.
- The customer profile page should display online test progress (test name, time taken, score, status), personal information, driver notes with lesson times and comments, special needs, and driver and student photos, based on Liam's interface sketch.
- The secretary needs an interface to look up customers and manage reservations on their behalf.
- The IT officer needs an admin panel to manage user accounts, reset passwords, and block access.

  The owner needs access to activity reports and the ability to enable or disable packages

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Users have internet access
- Customers are comfortable using web/mobile apps
- DMV provides updates digitally
- Payment processing is handled securely via third-party integration.
- Each driving lesson is exactly two hours long
- Pickup and drop-off locations for driving lessons are the same

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The system depends totally on internet connection
- It depends on DMV updates
- Limited customization without a developer being involved.
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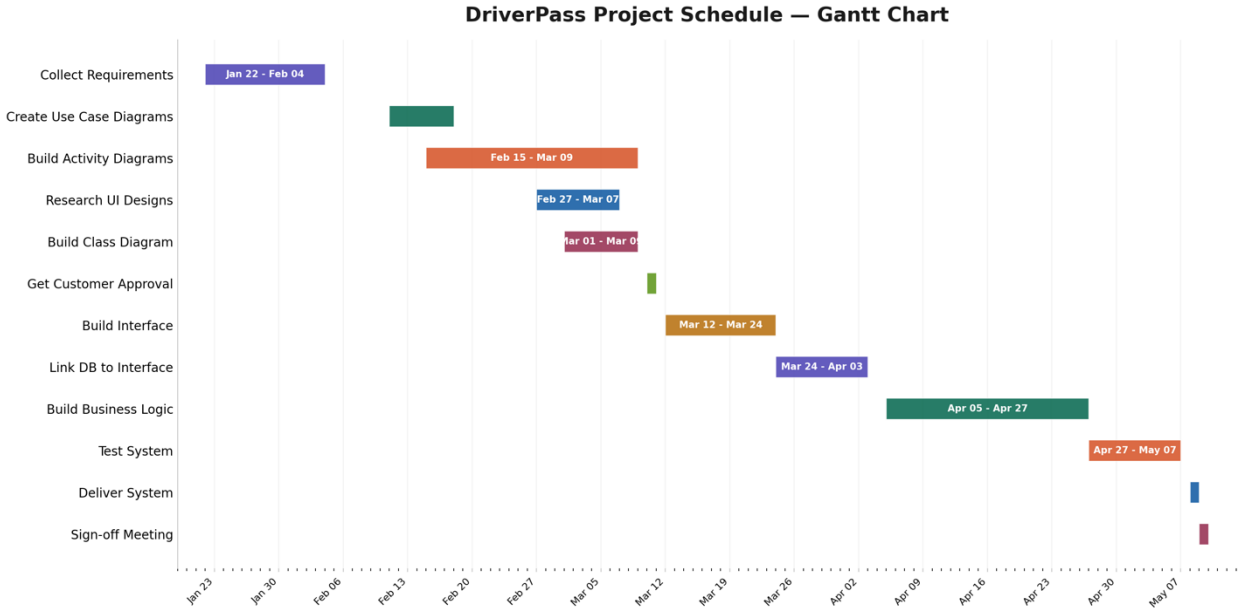
Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.


DriverPass Project Schedule — Gantt Chart
Collect Requirements
Create Use Case Diagrams
Build Activity Diagrams
Research UI Designs
Build Class Diagram
Get Customer Approval
Build Interface
Link DB to Interface
Build Business Logic
Test System
Deliver System
Sign-off Meeting
Jan 22 - Feb 04
Feb 15 - Mar 09
Feb 27 - Mar 07
Mar 12 - Mar 24
Mar 24 - Apr 03
Apr 05 - Apr 27
Apr 27 - May 07
Jan 23
Jan 30
Feb 06
Feb 13
Feb 20
Feb 27
Mar 05
Mar 12
Mar 19
Mar 26
Apr 02
Apr 09
Apr 16
Apr 23
Apr 30
May 07